

Prehistory, History, Present, Future

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Prehistory | Origins of the Internet

- The desire to communicate (also remote)
- First instruments: language and speech
- Communication form: written



Writing on rocks, cave walls, papyrus 😊

- Remote communication?
 - Fire signals
 - Morse Code
- 19th century: electricity and telegraph

History | Origins of the Internet

1837 Charles Babbage designs “first computer”



1863 Jules Verne:

"photo-telegraphy allowed any writing, signature or illustration to be sent faraway - every house was wired"

1891 – Ada Lovelace – first program for Babbage machinery

1936 Alan Turing – describes Turing machine

1939 HP is founded (Bill Hewlett, David Packard)

History | Origins of the Internet

1961-1972: first communication's attempts using packet-switching

1961: Kleinrock – proposed a theoretical model

1964: Baran – implemented the communication among US military computers

1967: ARPAnet was projected by Advanced Research Projects Agency

1969: first operational node ARPAnet, a network formed by 4 computers

History | Origins of the Internet

1972:

- public demonstration of ARPAnet technologies
- NCP (Network Control Protocol) – the first host-host protocol
- First program for electronic mail (e-mail)
- The sign @ is introduced
- ARPAnet contains 15 nodes

History | Origins of the Internet

1972-1980: The *Internetworking* concept appeared. Also, proprietary networks appeared.

1973: DARPA (Defense Advanced Research Projects Agency) – interconnected networks; Robert Metcalf (Harvard) developed Ethernet technology that allowed data transfer using coaxial cable

1974: Cerf si Kahn – proposed a communication protocol entitled TCP(Transmission Control Protocol)

1978: TCP/IP protocols stack was standardized via RFC (Request For Comments) documents

In the late of 70s: proprietary networks stacks appeared: DECnet, SNA, XNA

1979: ARPAnet contained 200 nodes

History | Origins of the Internet

1980-1990: new protocols, the network number was increasing, Internet

1983: TCP/IP was used

1982: SMTP (Simple Mail Transfer Protocol) was defined

1983: DNS (translation of host name into IP address and vice versa) appeared

1985: FTP (File Transfer Protocol) protocol appeared

1986: Internet backbone appeared

1988: some congestion control mechanisms for TCP were introduced

History & Present | Origins of the Internet

1990 ->2000->2016: World Wide Web, new commercial network applications

Early '90:

- World Wide Web**

After the late '90 -> present :

- Global Networks
- Wireless networks
- Commercial applications of the Internet

instant messaging, P2P file sharing
social networks, video-streaming,
cloud-computing,

...

quantum internet prototypes (Europa, US), Micius – satellit

quantum

Present

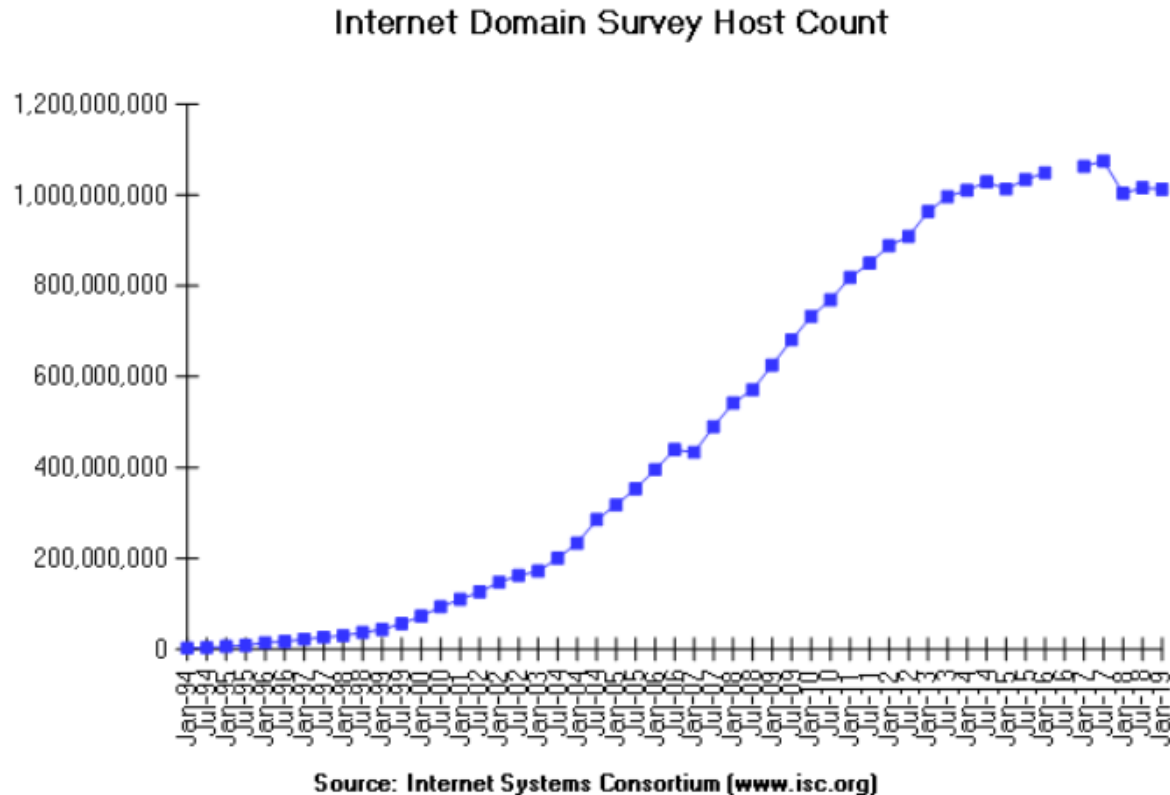


Figure. Hosts number form January 1994 till January 2019

Source: <https://www.isc.org/network/survey/>

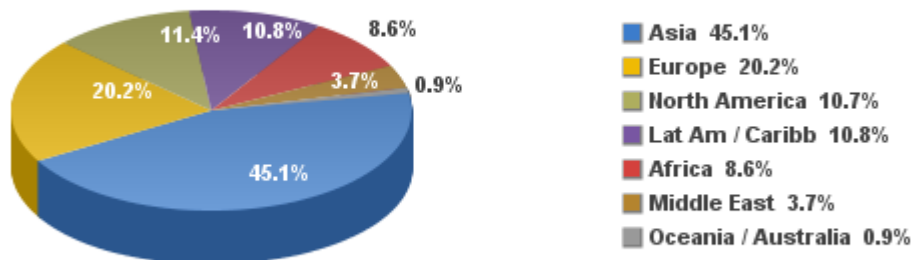
Past

WORLD INTERNET USAGE AND POPULATION STATISTICS December 31, 2013

World Regions	Population (2014 Est.)	Internet Users Dec. 31, 2000	Internet Users Latest Data	Penetration (% Population)	Growth 2000-2014
Africa	1,125,721,038	4,514,400	240,146,482	21.3 %	5,219.6 %
Asia	3,996,408,007	114,304,000	1,265,143,702	31.7 %	1,006.8 %
Europe	825,802,657	105,096,093	566,261,317	68.6 %	438.8 %
Middle East	231,062,860	3,284,800	103,829,614	44.9 %	3,060.9 %
North America	353,860,227	108,096,800	300,287,577	84.9 %	177.8 %
Latin America / Caribbean	612,279,181	18,068,919	302,006,016	49.3 %	1,571.4 %
Oceania / Australia	36,724,649	7,620,480	24,804,226	67.5 %	225.5 %
WORLD TOTAL	7,181,858,619	360,985,492	2,802,478,934	39.0 %	676.3 %

Tabel. INTERNET
USAGE STATISTICS
- The Internet Big
Picture

Internet Users in the World Distribution by World Regions - 2013 Q4



Source: Internet World Stats - www.internetworldstats.com/stats.htm
Basis: 2,802,478,934 Internet users on Dec 31, 2013
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Figura. Internet Users in the
World by Regions
December 2013 | Source:
<http://www.internetworldstats.com/>

Present

**WORLD INTERNET USAGE AND POPULATION STATISTICS
2022 Year Estimates**

Table. Internet usage statistics relative to the number of people in the main geographic areas of the world

World Regions	Population (2022 Est.)	Population % of World	Internet Users 30 June 2022	Penetration Rate (% Pop.)	Growth 2000-2022	Internet World %
Africa	1,394,588,547	17.6 %	652,865,628	46.8 %	14,362 %	11.9 %
Asia	4,352,169,960	54.9 %	2,934,186,678	67.4 %	2,467 %	53.6 %
Europe	837,472,045	10.6 %	750,045,495	89.6 %	614 %	13.7 %
Latin America / Carib.	664,099,841	8.4 %	543,396,621	81.8 %	2,907 %	9.9 %
North America	374,226,482	4.7 %	349,572,583	93.4 %	223 %	6.4 %
Middle East	268,302,801	3.4 %	211,796,760	78.9 %	6,378 %	3.9 %
Oceania / Australia	43,602,955	0.5 %	31,191,971	71.5 %	309 %	0.6 %
WORLD TOTAL	7,934,462,631	100.0 %	5,473,055,736	69.0 %	1,416 %	100.0 %

**Internet Users in the World
by Geographic Regions - 2022**

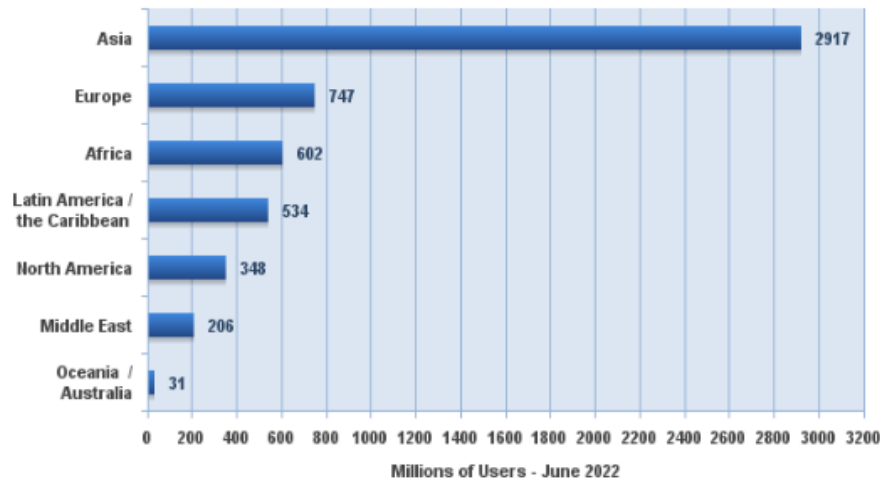


Figure. Internet users in the world - geographically divided
June 2022| Source:
<http://www.internetworldstats.com/>

Source: Internet World Stats - www.internetworldstats.com/stats.htm
Basis: 5,385,798,406 Internet users estimated in June 30, 2022
Copyright © 2022, Miniwatts Marketing Group

Present

Number of internet users worldwide as of 2022, by region
(in millions)

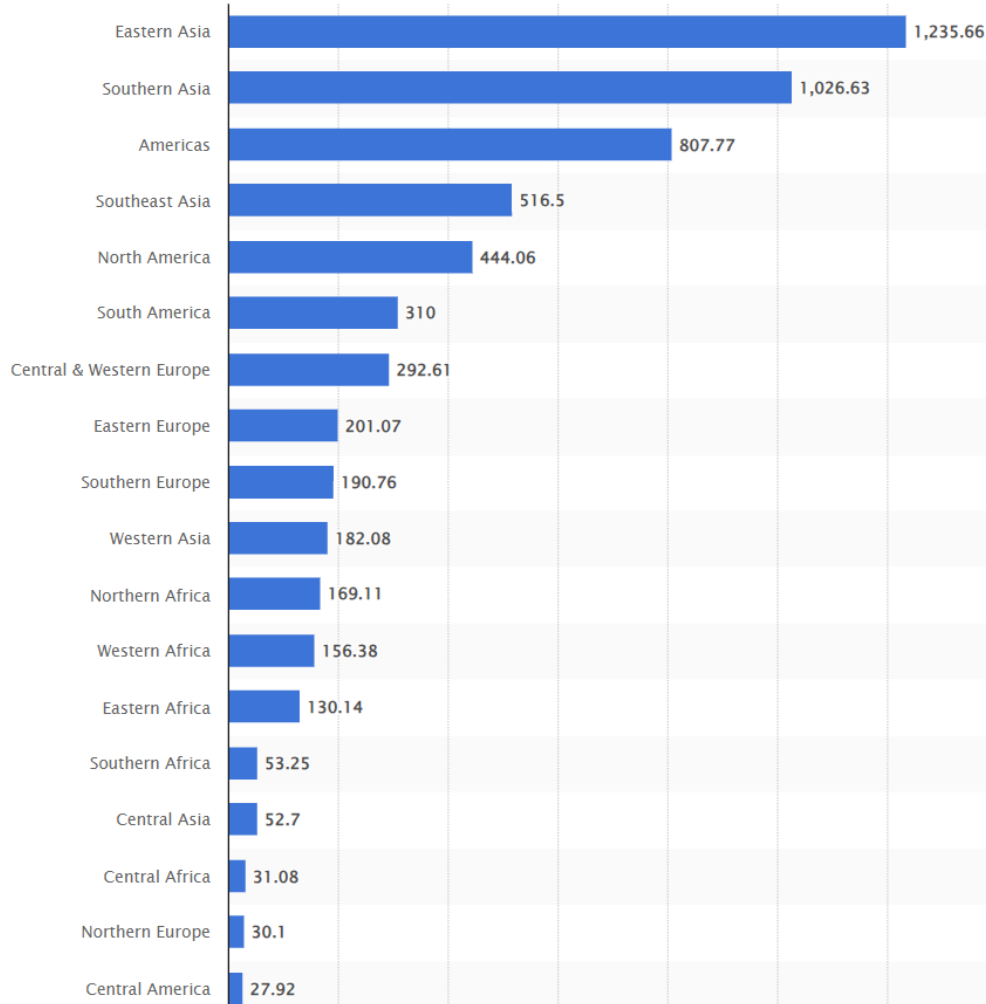


Figure. Internet users in the world
- geographically divided
June 2022| Source:
<https://www.statista.com/statistics/249562/number-of-worldwide-internet-users-by-region/>

Prezent

Number of internet users worldwide as of 2023, by subregion
(in millions)

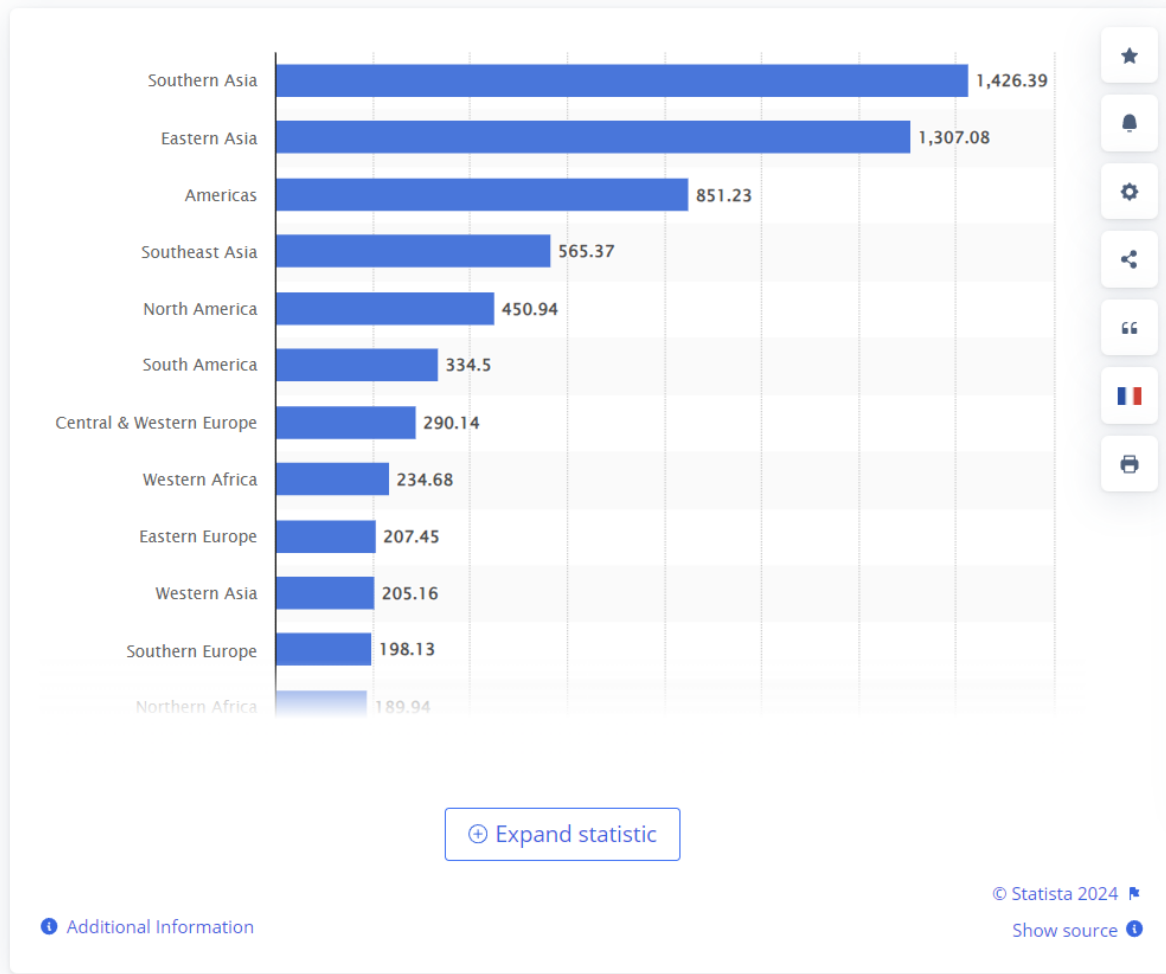
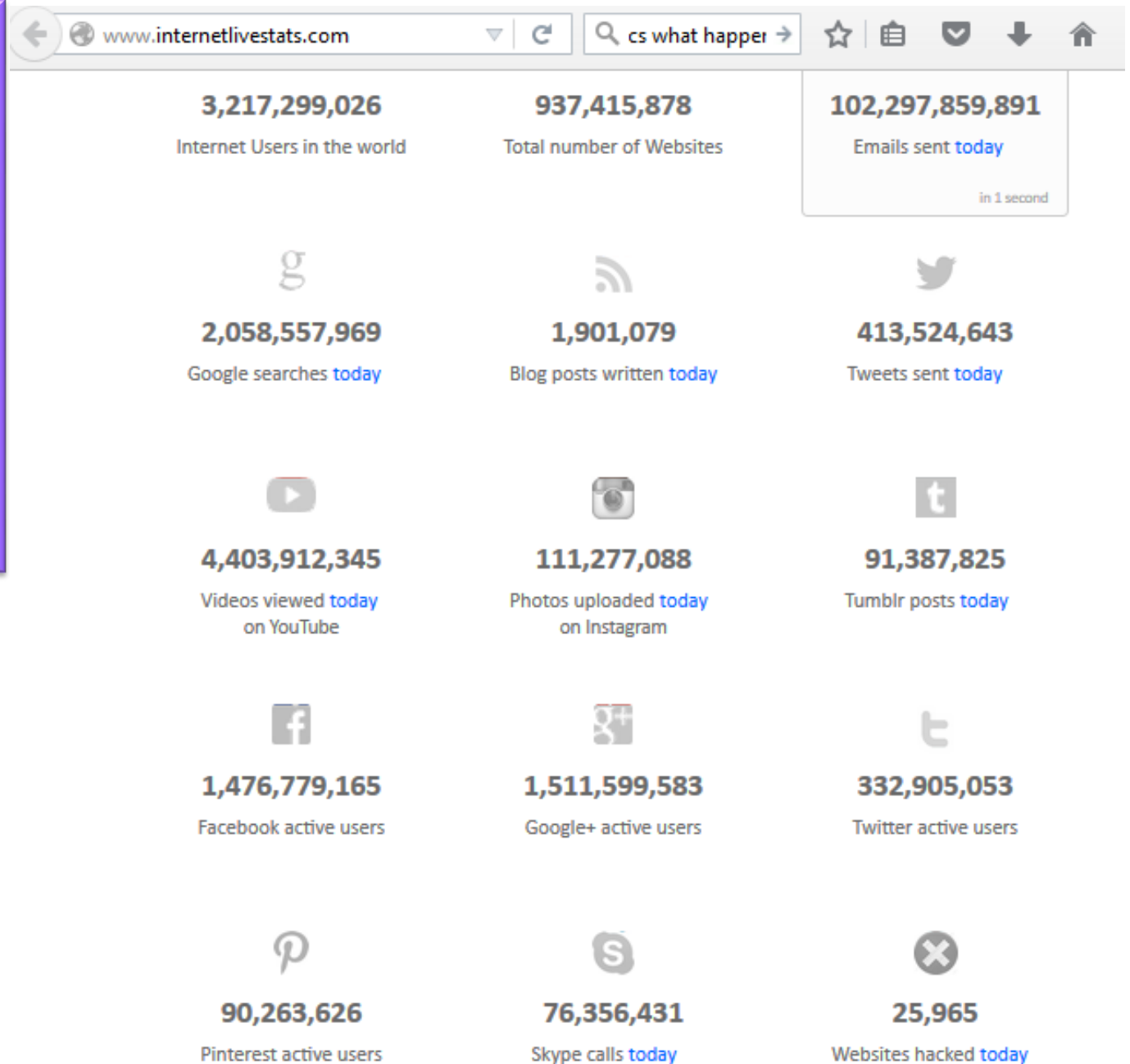
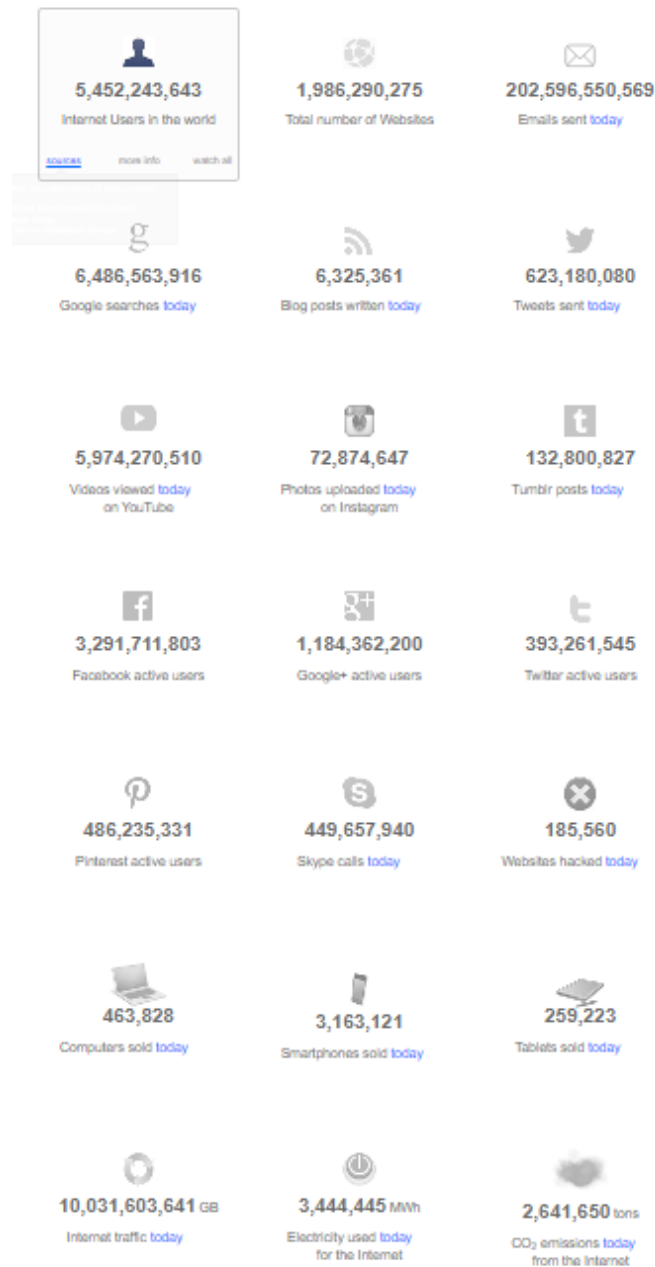


Figure. Internet users in the world
- geographically divided
June 2024| Source:
<https://www.statista.com/statistics/249562/number-of-worldwide-internet-users-by-region/>

2015



2022



Internet Live Stats has been cited by



The Official World Wide Web Anniversary Site
World Wide Web Consortium (W3C)
World Wide Web Foundation

Present

Computer Network course – Weeks 1-13-

The future

Computer Network course – Week14-

Resources

- Internet Society: <http://www.isoc.org>
- FNC – Federal Networking Council:
http://www.nitrd.gov/fnc/Internet_res.html
- InternetWorld Stats: <http://www.internetworldstats.com>
- Internet Systems Consortium: <http://www.isc.org>
- Internet Assigned Numbers Authority (IANA): <http://www.iana.org/>
- Cisco Systems: <http://www.cisco.com>
- <http://www.nethistory.info>
- <http://www.caida.org/>
- http://news.netcraft.com/archives/web_server_survey.html
- <http://www.swivel.com/>
- <https://www.facebook.com/cisconetworkingacademy> :)
- <http://www.internetlivestats.com/>
- <http://www.worldometers.info/>
- <http://www.internetsociety.org/globalinternetreport/?gclid=CN6a2Y2wqMgCFUfkWgodZ8gL9A>
- <https://www.coursera.org/learn/quantum-computing-for-everyone-an-introduction>